ADVANCED THEORETICAL SYNTHESIS

The Static Gravitational Well Framework

A complete cosmological alternative replacing expansion, dark energy, and dark matter with a bounded, resonant macro-cavity.

The Paradigm Shift

Moving from expanding spacetime to a stationary bounded resonator.

Standard ΛCDM Paradigm

Expanding universe driven by dark energy, requiring unobserved dark matter halos and a hot Big Bang temporal origin.

Static Cavity Framework

A 134.1 Gly bounded resonant metric shell structure where scaling profile $S(r)$ governs dynamics natively.

Gravitational Wave Propagation

Replacing expanding-space luminosity distance with proper metric attenuation.

Tensor Perturbations

Waves traverse background $S(r)$ without spatial expansion.

Proper Volume Scaling

Spatial volume scale $V \propto S(r)^3$ governs wave energy.

Strain Attenuation

Amplitude scales as $h \propto 1 / (S(r)^2 r)$ matching LIGO.

Covariant Action Principle

Deriving static metric potentials from modified field equations.

Scalar-Tensor Cavity Action

Incorporates a preferred-frame temporal scalar field ϕ whose vacuum expectation value sets the stationary scaling profile $S(r)$.

Modified Field Equations

$G_{\mu\nu} = 8\pi G_0 (T_{\mu\nu}^{\text{matter}} + T_{\mu\nu}^{(\phi)})$ yielding stable Dirichlet boundary solutions at R_{edge}.

Steady-State Primordial BBN

Replacing temporary cooling plasmas with permanent thermodynamic eigenmodes.

Core Thermal Environs

Inner galactic metric wells maintain fusion-level temperatures ($S \sim 1$).

Metric Temperature Gradient

Local thermal eigenmode follows $T(r) = T_0 / S(r)$ across concentric shells.

Equilibrium Abundances

Helium-4 ($\sim 25\%$ mass fraction) maintained by continuous recycling.

Static Cosmic Structure ($f\sigma_8$)

Replacing chronological collapse with spatial eigenmode density settling.

Standing Wave Resonator

Matter naturally segregates into nodes and antinodes of the master metric potential $S(r)$ without requiring 13.8 billion years of growth.

Spatial Covariance (σ_8)

Measures spatial covariance of metric eigenmode density fluctuations across intermediate shells, reproducing observed survey clustering.

COMPLETE FRAMEWORK SYNTHESIS

A Fully Airtight Static Universe

Unified macro-cosmology, galactic dynamics, chemistry, particle physics, and gravitational waves under one geometric framework.

Ready for Empirical Testing